

Summary	Practiced software engineer, enthusiastic learner, and intuitive communicator with 14 years of professional programming experience in a broad variety of demanding technical environments	
Skills + Languages	Full-Stack Software Engineering, Data Science, Data Engineering, Cloud Architecture, DevOps, MLOps, Product Management Java, Clojure, TypeScript, Python, Go, Rust, C#, C, Lua, Erlang, Haskell, OCaml	
Experience	Principal Software Engineer — Wolfram Research	2023-Current
	+ Inherited and led the rearchitecture and migration of the Wolfram Cloud application to Kubernetes + Developed and deployed back-end services in Java, Typescript, and Wolfram Language related to kernel management, expression evaluation, server-side rendering, and Mathematica notebook state management + Addressed synchronization bugs and failure modes in legacy Java code	
	Founder, CTO, Principal Software Engineer — NuID [nuid.io]	2017-2022
	+ Designed and implemented an elastic, highly available authentication API and web-based developer portal using Clojure, TypeScript, AWS, and Terraform + Designed and implemented a cloud-native developer experience within AWS enabling on-demand development environments, stateful integration testing, live previews, progressive delivery, and structured observability + Led development and maintenance of open-source front-end and back-end libraries in Java, JavaScript, Go, Ruby, and Clojure, helping hundreds of developers integrate our authentication API into their projects + Hired and led small, distributed teams across engineering and product development which were responsible for onboarding thousands of users onto our authentication service and public key infrastructure + Collaborated with professor of cryptography and ACM Fellow Jonathan Katz to develop a zero-knowledge authentication protocol for which we were granted patents US 10880089, 11496310	
	Data Scientist — Microsoft	2015-2017
Education	+ Implemented and maintained pipelines in Spark and Hadoop-like architectures driving dashboards and analyses relied upon by senior leadership and the CTO + Led engineering and deployment of a machine learning model for device age, informing high-level Windows 10 marketing and delivery strategies across the Windows organization + Designed and implemented a compiler and toolchain in C# for describing compute workflows that improved metric accuracy, reliability, and compute efficiency across our team	
	Software Engineer — Rustici Software	2013-2015
	+ Contributed to open-source eLearning libraries and browser extensions which increased adoption of an emerging eLearning specification and established new revenue streams for our team + Designed and implemented an open-source conformance suite for testing new and existing specification implementations, leading to an ecosystem-wide improvement in compatibility and improving the visibility of our products	
	M.S. Computer Science — Vanderbilt University	2013-2015
	+ Thesis: Design Patterns in Concurrent and Asynchronous Systems + Project: Distributed Online Record Linkage with MinHash and Storm + GPA: 4.0/4.0	
References	B.S. Computer Science — Vanderbilt University	2011-2015
	+ Completed B.S. and M.S. in 4 years + Fine art minor focusing in drawing and printmaking + GPA: Cumulative 3.777/4.0 - Major 3.854/4.0	
	J.D. Hollis — jd@theconsultingcto.com	
	+ Collaborator on infrastructure at NuID	
	Pallavi Baral — pallavibaral@gmail.com	
Interests	+ Manager at Microsoft	
	Hiking, Printmaking, Programming Languages, Game Design, Mereology, Music	